F-23A Specter

Introduction

On June 22nd, 1990, one of the most unique fighter designs was rolled out of the Northrop test facility hangers at Edwards Air Force base. This aircraft, with its gigantic V-tails and diamond shaped wing had an almost alien appearance. The primary competitor, the YF-22 took a much more traditional approach with conventional vertical and horizontal stabilizers and look more or less like a “super F-15”.

The YF-23 took to the air the following August and took part in the Advanced Tactical Fighter (ATF) Demonstration/Validation (DemVal) phase for the ensuing three months. After the conclusion of the DemVal phase, in April 1991 the Lockheed entry was selected as the winner and went on to develop the F-22 Raptor. Sadly, this unique fighter design was relegated to the history books as the two prototypes were placed in storage and eventually donated to museums.

Interest in the F-23 design persisted over the years in aviation enthusiast circles. Like most prototypes, the YF-23 was not the final design for a production airplane. As was seen in the YF-22, the design was to evolve considerably to the final production configuration. In the aviation press there were some limited artwork and concept drawings during the ATF DemVal period, but no detailed renderings were available.

Late in the DemVal period, Northrop had released unclassified scale drawings of their production configuration in the form of General Arrangement Plans. These plans detailed some of the equipment, locations of major subsystems and all of the major dimensions of the vehicle to scale. These plans have resurfaced in the past decade or so and have been publicly available in high resolution.

It is with these plans that I formed the basis of creating a “mod” for DCS World. There were some previous attempts to model the production configuration, but they were inaccurate in both shape and details. As a result, I commissioned artwork with tight adherence to these plans to ensure that I had captured every detail and shape that is detailed in those general arrangement plans.

The external model was fairly straightforward as I had detailed plans, but the cockpit was far more challenging as there are almost no references for it. However, there were *some* refences. The best refence was a cross-section illustration in one of the sheets of the general arrangement plans that shows the instrument panel and HUD shape in a fair amount of detail.

Combined with artwork from Aldo Spadoni, an artist who worked as avionics engineer for Northrop and provided some official artwork, I was able to arrive at what I believe is an accurate representation of the instrument panel. Having the instrument panel is a big step but that still left me with quite a bit of the cockpit left to create essentially from scratch.

I approached this by noting how the F-22’s side panels were arranged and cross-referencing how the later McDonnell Douglas ergonomic practices are implemented in the F-15E. The F-22, for obvious reasons, makes for a good reference because it had to meet the same requirements that the F-23A would have had with high levels of automation. So, the side panels and flight controls (stick and throttles) are a cross between the F-22 and F-15E.

It took about 18 months to complete the artwork but that is but part of the challenge of creating a mod for DCS World. Since then, I have brought in several new members to help me bring the F-23 to life. For our initial release, we are releasing a Simplified Flight Model (SFM) in what we call F-23A Block 20 increment 1. As we update the SFM version, we will change the increment number to reflect that particular release.

We are also planning on moving to an Enhanced Flight Model (EFM). This will allow for better flight model tuning and better flying qualities. This version will be called the F-23A Block 30. It is possible that we may iterate further such as a standalone version or other enhancements, however, we do not have firm plans for those at this state.

The changes Eagle Dynamics made to DCS World have set us back longer than we would have preferred but now that we have worked around them and are excited to bring to you this mod. It is a true passion project and something I personally have dreamt about for literally decades.

I truly hope you enjoy it!

Smash

Installation & Operation

Requirements: DCS World and either the F-15C or the Flaming Cliffs 3 pack.

Installation: The F-23A Block 20 installs like most other DCS World mods:

1. Navigate to your DCS World saved games folder: \Users\[Your Username]\Saved Games\DCS\
2. If you don't already have a "Mods" folder, create one in the DCS directory
3. Inside the Mods folder, create an "Aircraft" folder (if it doesn't exist)
4. Drop the entire F-23A folder into the Aircraft directory

Your final path should look like: DCS\Mods\Aircraft\F-23A\

Important Setup

Before flying the F-23, you **must first spawn into an F-15C** once per DCS session. This only needs to be done once when you start DCS World—not for every mission or F-23 spawn afterward.

Recommended process:

1. Start DCS World and spawn into any F-15C mission
2. Exit the mission or switch aircraft to the F-23

3. You can now freely fly the F-23 for the remainder of your session

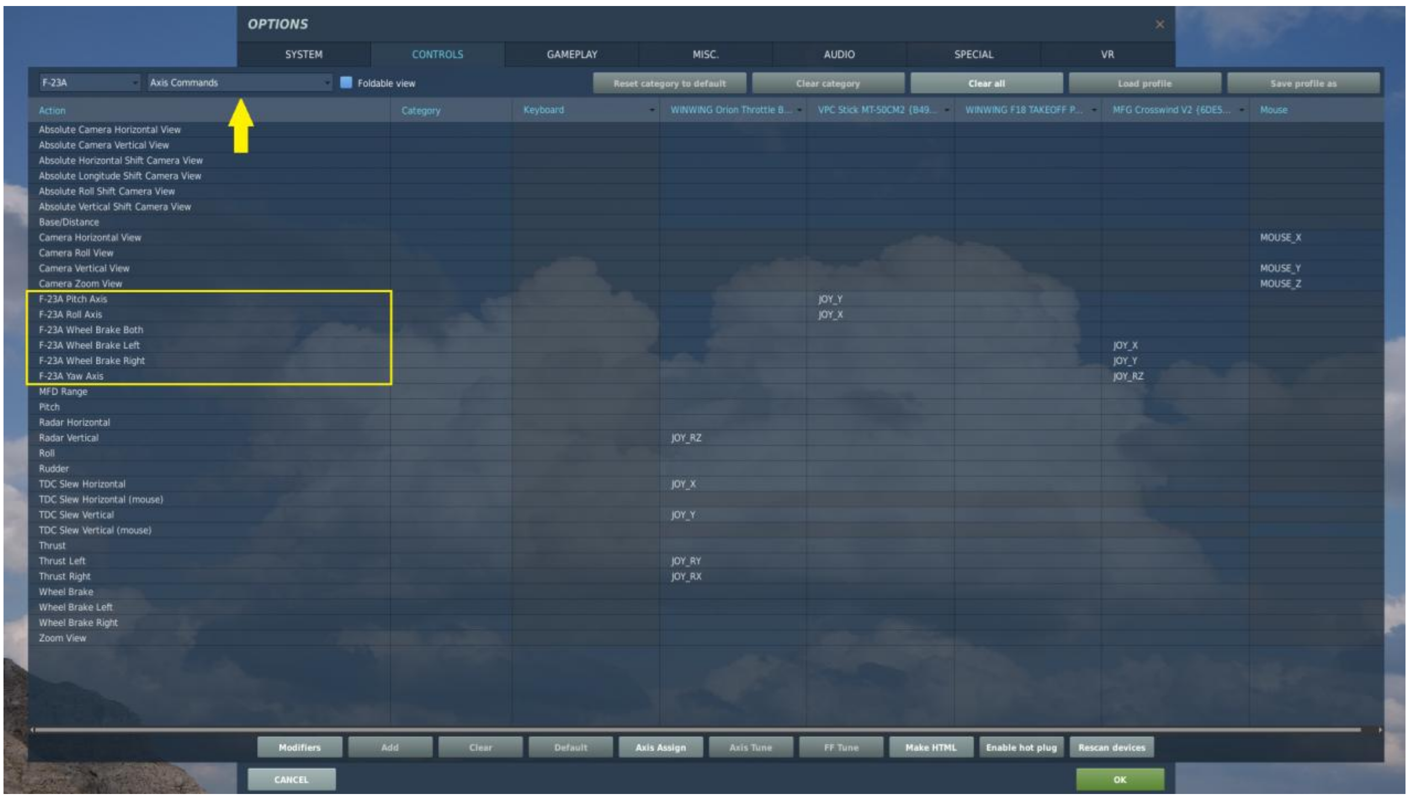
Note: You'll need to repeat this F-15C spawn step each time you restart DCS World.

Setup and Key Binds

Key Binding Setup

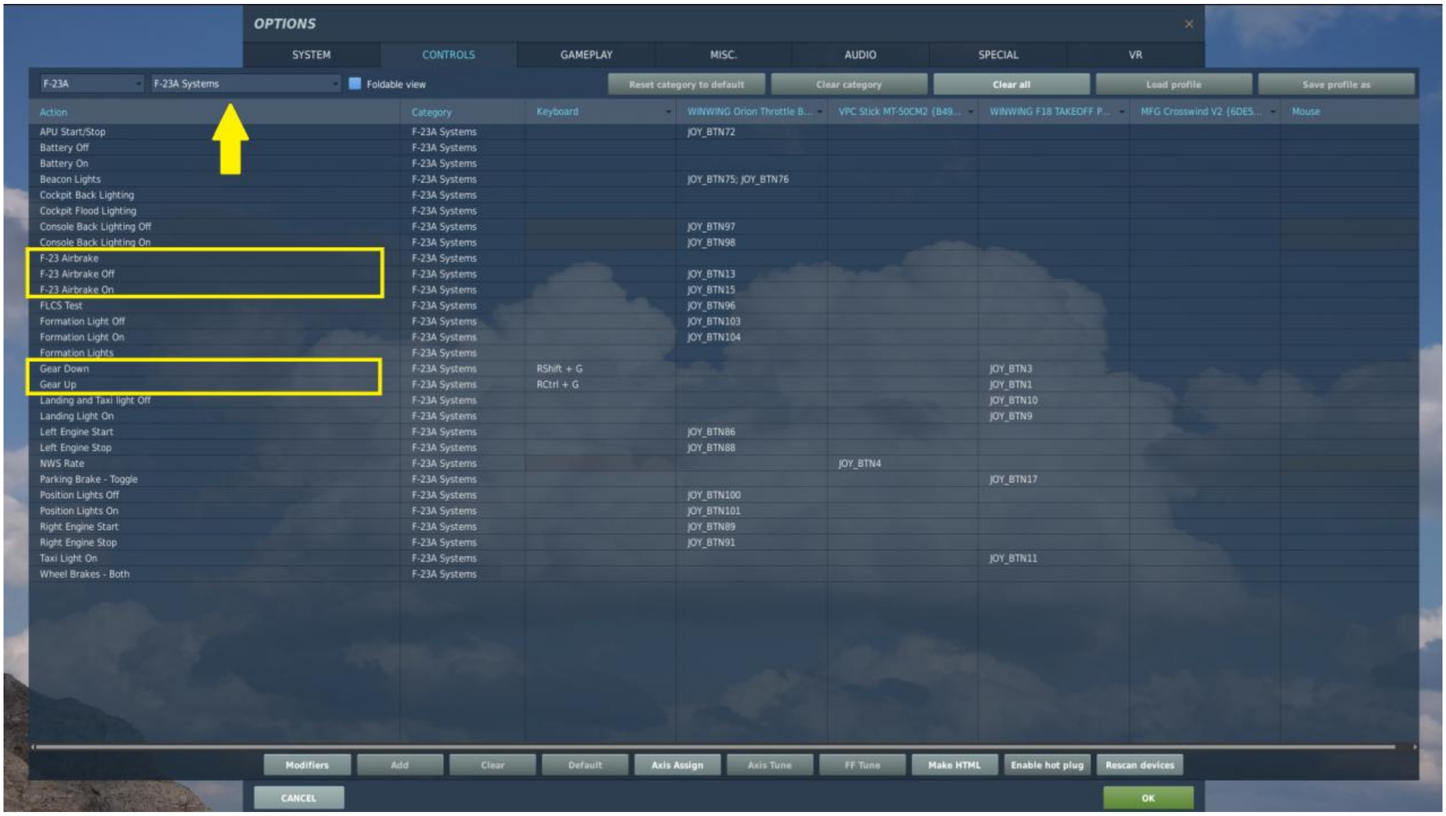
Like many mods, the F-23 requires specific key bindings for proper operation. **You must use the F-23-specific binds rather than default DCS controls.**

Flight Controls (Axis Commands Tab) The F-23 has dedicated axis commands for all primary flight controls. Look for bindings labeled "F-23A" followed by the control name (e.g., "F-23A Pitch Axis"). While you may see what appear to be duplicate bindings, always select the ones specifically labeled for the F-23A.



Critical System Binds (F-23A Systems Tab) Several essential systems require F-23-specific bindings:

- **Landing Gear:** You must bind "F-23A Gear Up" and "F-23A Gear Down." Default DCS gear controls will not work.
- **Air Brakes:** Use F-23-specific air brake bindings, as default controls are incompatible.

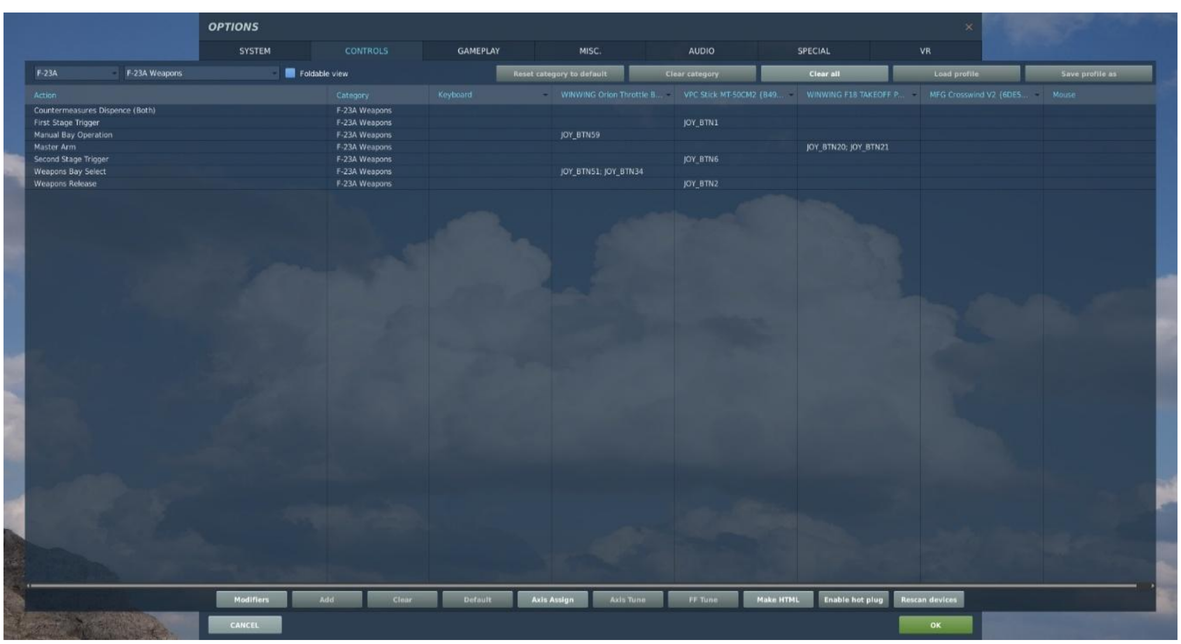


Weapons Systems Weapons operations require specific F-23 bindings for proper functionality:

Weapon Bay Operations: The F-23 has two weapon bays—forward (typically AIM-9s) and main. The forward bay must be manually selected when using AIM-9s. **Important:** If AIM-9s are selected but the forward bay is not opened, missiles will fire through the aircraft nose and collide with the collision shell, resulting in lost shots.

Triggers and Release:

- **Weapon Release:** Opens selected weapon bay doors when held, allowing missile launch
- **Gun Trigger:** Two-stage operation—first stage opens the gun muzzle door; second stage fires the gun.



F-23A Quick Start Guide

The F-23A features a streamlined startup sequence that reflects the simplified systems integration typical of fifth-generation fighter aircraft. Much like the real-world F-22 Raptor, the F-23A's startup procedure is remarkably straightforward, designed to minimize pilot workload and reduce time from cold aircraft to mission-ready status.

Step-by-Step Startup

1. Master Battery Power

- Switch **MASTER BATTERY** to **ON**
- Verify electrical power is available to essential systems

2. Auxiliary Power Unit (APU)

- Switch **APU** to **START**
- Wait approximately **5 seconds** for APU to spool up and stabilize
- APU provides electrical power and pneumatic pressure for engine start sequence

3. Engine Driven Generators

- Switch **LEFT ENGINE GENERATOR** to **ON**
- Switch **RIGHT ENGINE GENERATOR** to **ON**
- Generators will come online automatically once engines are started

4. Engine Start

- **LEFT ENGINE START** - Switch to **START**
- **RIGHT ENGINE START** - Switch to **START**
- Monitor engine parameters during startup sequence on the Up Front Display Controler (UFCD)

5. Systems Check

- Verify **Up Front Displays (UFDs)** are clear of cautions and warnings
- Check all engine parameters are within normal operating ranges
- Confirm flight control system initialization is complete

6. Weapons Bay Configuration

- Close **WEAPONS BAY DOORS**
- Verify **WP BAY** annunciator exinquishes on right UFD

F-23A DP231

The Northrop airplane that is most commonly associated with the ATF program is the YF-23, the prototype. Northrop and McDonnell Douglas built two Prototype Air Vehicles (PAV) to support the Dem/Val phase of the program. They were not the proposed production configuration but were close enough to validate aerodynamic performance and systems predictions by the contractors.

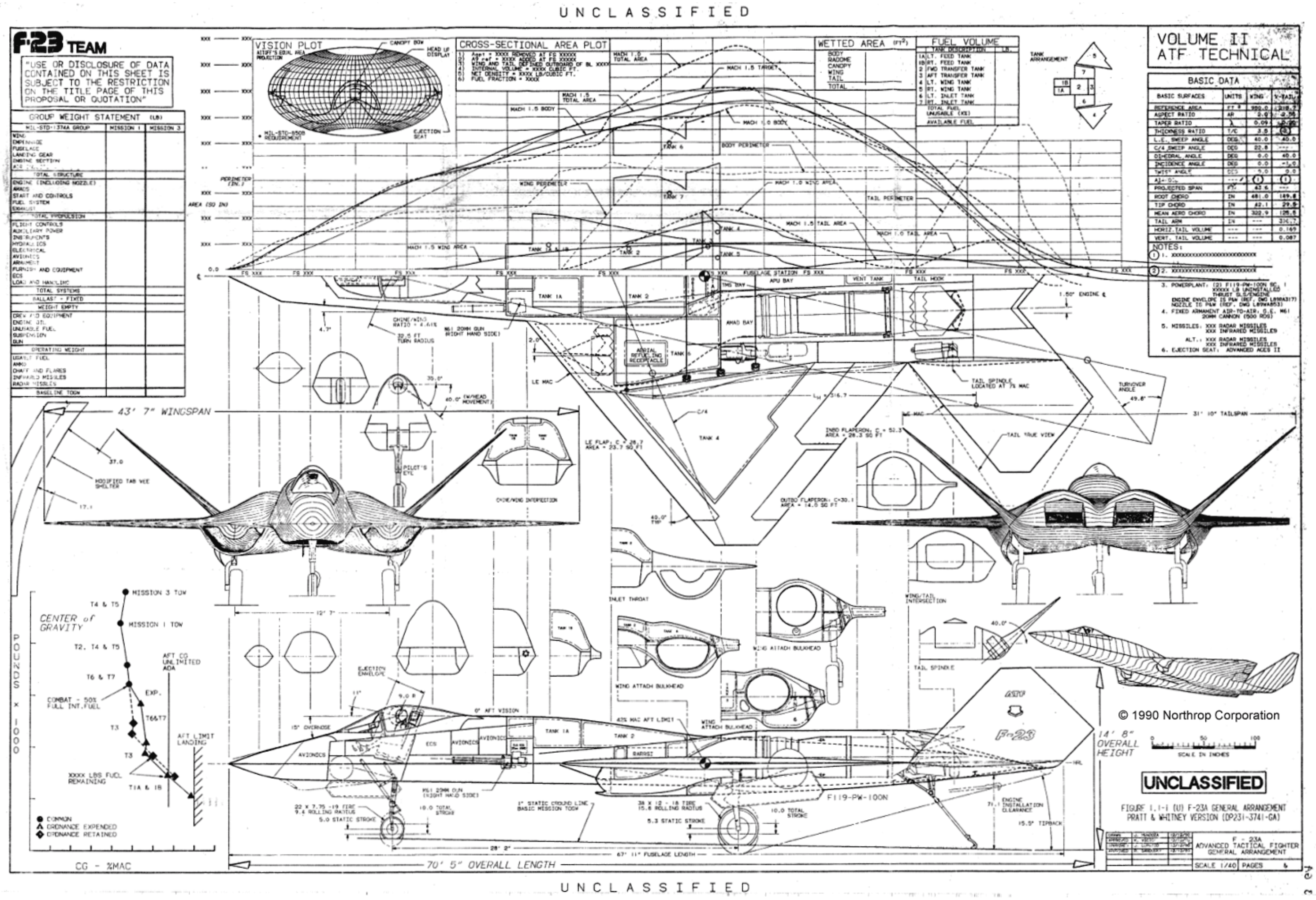
Like the F-22 Raptor, the F-23 team continued to iterate on the production proposal. The USAF was not fond of a single weapons bay solution that was originally proposed by Northrop so the design evolved by stretching the forward fuselage which would allow for a second weapon bay as well as more fuel and internal volume for avionics and subsystems. Additional changes included bringing the engines closer together and refined planform and outer mold line shaping to further enhance the vehicle's low observable (LO) signature.

This configuration now had a smaller, less voluminous main weapons bay located between the inlets and now a smaller forward weapons bay behind the nose wheel gear bay. The latter bay was sized just large enough to accommodate two AIM-9M missiles. Additionally, there was an M61 20mm gun located on the right forward fuselage just behind the cockpit.

While we accurately depicted these important details, one critical piece of information that is lacking in these General Arrangement Plans is what launcher technology. There is no detection of what launcher solutions would have been employed in either weapon bays. The only limited concrete information available publicly at the moment is that the main weapons bay would not have had door mounted launch rails nor would it have had the "Stores Magazine and Launch System" that Northrop patented in 1987.

We do know that the forward weapon bay would have had a form of a trapeze launcher that would extend the missile out of the bay to allow the seeker a clear field of view to acquire the target. The exact configuration of this launcher is not known so the best reference is the LAU-141 launcher for the F-22 Raptor. This works but it is unlikely that the F-23A would have had a similar design as this launcher as this approach brings the missile down and forward which brings the nose of the missile close to the nose gear strut. This approach would still be viable, though not optimal.

For the main weapons bay, we had such limited information on launcher technology we elected to use the Northrop launcher as detailed in the 1987 patent. What is satisfying about this hypothetical arrangement is that it realistically allows for 6 AIM-120C AMRAAMs to be carried in the main weapons bay, matching the capacity of the F-22 Raptor.



Sheet 1 of the General Arrangement Plans for Design Proposal 231

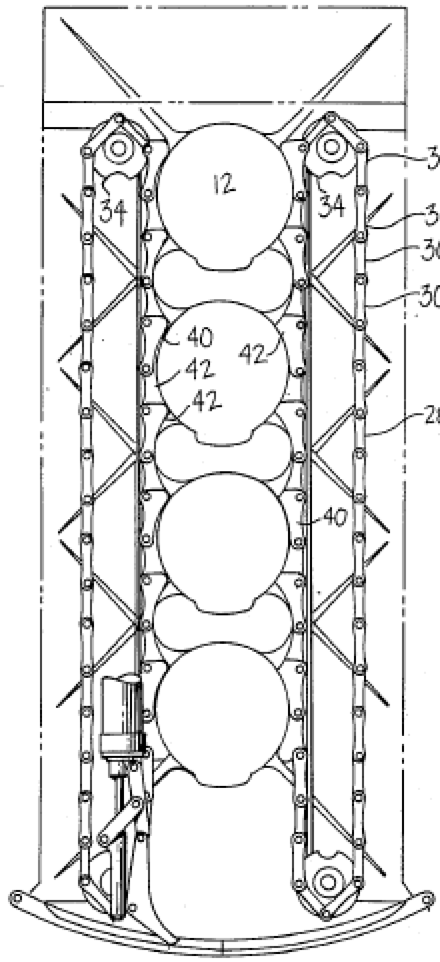


Fig. 3

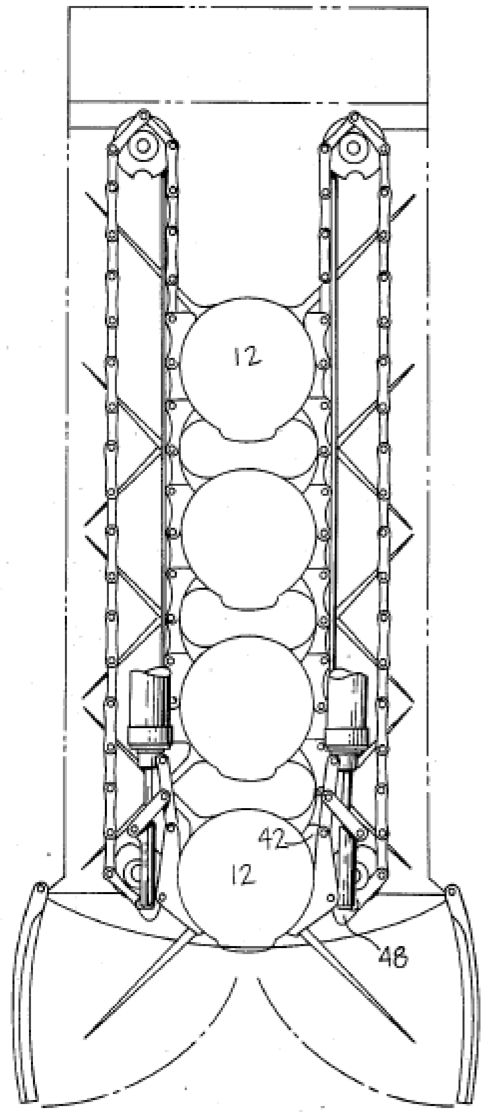


Fig. 4

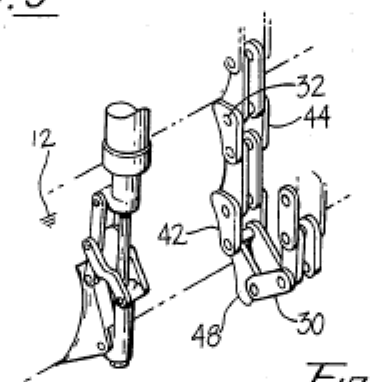


Fig. 5

Drawings from Northrop patent # 4,702,145 showing basic missile stowage and launcher configuration

Flight Modeling

Developing the flight model for the F-23A presented a unique set of challenges that went far beyond the typical difficulties encountered when modeling existing aircraft. As the F-23A production configuration never flew, and the performance details of the YF-23 prototypes remain largely classified, we had very little concrete data to draw upon for reference. What is publicly known provides only tantalizing glimpses of the aircraft's capabilities: the YF-23 demonstrated low speed and high angle of attack control that exceeded what was possible with contemporary fighters like the F-15, and it was one of the hallmarks of the Advanced Tactical Fighter (ATF) program that both contenders had to demonstrate supercruise capability. Historical accounts suggest that the YF-23 not only met but exceeded the ATF supercruise specifications, giving us at least a performance baseline to work toward.

The Simplified Flight Model Challenge

The decision to initially develop the F-23A using DCS World's Simplified Flight Model (SFM) was driven by practical considerations, but it came with significant technical limitations that required creative solutions. The SFM, while more accessible for modding than the Enhanced Flight Model (EFM), operates on a fundamentally simplified approach to aircraft performance modeling. Unlike real-world flight dynamics where performance characteristics flow naturally from aerodynamic properties, mass distribution, and control authority, the SFM requires developers to essentially "paint" performance envelopes across the flight regime using a series of interconnected lookup tables and curves.

This approach creates inherent compromises that became apparent early in our development process. When tuning available G-loading at a given Mach number, for example, increasing performance at one altitude often resulted in unrealistic performance penalties at others. The SFM's simplified handling of compressibility effects, Reynolds number variations, and altitude-dependent aerodynamic characteristics meant that achieving realistic performance across the entire flight envelope required extensive iteration and careful balancing of competing requirements.

Parametric Performance Modeling Approach

Given these constraints, we developed what essentially amounts to a parametric aircraft performance model that we then mapped onto the SFM framework as best we could. This process began with establishing our performance targets based on the limited but credible data available about both YF-23 prototypes and the production F-23A requirements. We used the publicly available performance figures for the F-22A as a baseline, reasoning that both aircraft had to meet similar ATF requirements, and accounting for the documented performance differences between the YF-23 and YF-22 during the DemVal phase.

The parametric approach required us to make fundamental decisions about the aircraft's design philosophy that would drive our performance modeling. When trying to build the SFM for the F-23A, a critical question had to be asked: What is the goal of the SFM? The nature of the SFM's limitations meant we couldn't simply model realistic aerodynamics and let performance flow naturally from first principles. Instead, we had to prioritize certain flight regimes over others. Should we create a sea level Basic Fighter Maneuvers (BFM) monster optimized for knife-fight engagements, or focus on the high-altitude supercruising Beyond Visual Range (BVR) nightmare that the F-23A was designed to be?

Design Philosophy and Compromise

After extensive analysis of the ATF program requirements and the YF-23's documented strengths, the decision for our SFM was to focus primarily on high altitude performance while trying to sacrifice as little as possible at lower altitudes. This decision was driven by several factors: the F-23A's design emphasis on reduced radar cross-section and supercruise capability, the aircraft's large internal fuel capacity and efficient cruise characteristics, and the historical accounts suggesting that the YF-23's greatest advantages over the YF-22 were in the supercruise and stealth domains rather than pure agility.

This philosophy shaped every aspect of our performance modeling. We prioritized sustained turn rate over instantaneous turn rate, cruise efficiency over acceleration, and high-altitude maneuverability over sea-level agility. The extensive area ruling of the F-23A's fuselage, its large wing area, and the efficient airflow management provided by those distinctive diamond-shaped wings all pointed toward an aircraft optimized for high-speed, high-altitude operations.

Iterative Development Process

After several iterations of testing and tuning—a process that took the better part of eight months—what we accomplished was an SFM that provides reasonable agility across all altitudes and speeds, but where the F-23A truly shines above 36,000 feet. Each iteration involved extensive flight testing across the entire performance envelope, comparing our results against theoretical performance calculations and the limited benchmark data available from period sources.

The iterative process revealed some interesting characteristics that emerged naturally from our modeling approach. The F-23A's extensive focus on drag reduction, evident in every aspect of its design from the wing planform to the weapons bay configuration, created a unique challenge in the SFM framework. As so much of the aircraft's design philosophy centered on minimizing drag, we found that pilots would frequently need to utilize the speedbrake function when attempting to decelerate—a characteristic that actually aligns well with accounts of the YF-23's behavior during flight testing.

Performance Targets and Validation

The parametric modeling process began with establishing concrete performance targets based on our analysis of ATF requirements and YF-23 capabilities. **Chart 1** illustrates our flight envelope targets, showing the desired supercruise and maximum speed performance across the operational altitude range from sea level to 60,000 feet. The chart clearly demonstrates our focus on high-altitude performance, with peak supercruise capability of Mach 1.82 on military power at 36,000 feet, extending the YF-23's documented supercruise advantages into the production configuration.

The achieved performance curves in **Chart 1** show how closely we were able to match our targets within the SFM framework. Our F-23A achieves Mach 1.84 supercruise at 40,000 feet and a maximum speed of Mach 2.5 at 36,000 feet with afterburner—performance figures that reflect both the aircraft's exceptional high-altitude capabilities and the compromises necessary when working within the SFM's constraints. The slight variations between desired and achieved performance, particularly the enhanced supercruise capability at higher altitudes, actually emerged naturally from our drag modeling and represent what we believe would have been realistic performance improvements as the design matured from prototype to production.

Chart 2 presents the engine thrust characteristics that underpin these performance figures. The sea level thrust curves show peak afterburner thrust of nearly 80,000 pounds and military power thrust exceeding

53,000 pounds—figures that reflect the F119-class powerplant that would have equipped the production F-23A. The thrust curves reveal the characteristic behavior of modern fighter engines, with afterburner thrust remaining relatively constant through the transonic regime before beginning to decrease at higher Mach numbers, while military power thrust peaks around Mach 1.0 before declining as inlet efficiency and turbine inlet temperature limits are reached.

Maneuvering Performance and High-Altitude Optimization

The G-loading characteristics shown in **Chart 3** demonstrate both the achievements and limitations of our SFM approach. At sea level, our F-23A delivers up to 8.8G of available load factor, falling slightly short of the 9G target but remaining competitive with contemporary fighters. However, the chart reveals the aircraft's true strengths at higher altitudes: at 40,000 feet, the F-23A not only meets but exceeds the 9G target across much of the flight envelope, reaching 9.2G at Mach 1.8—a performance level that reflects the aircraft's large wing area and efficient high-altitude aerodynamics.

This high-altitude maneuvering advantage becomes even more apparent in the turn rate performance illustrated in **Chart 4**. While sea level turn rates are respectable, the F-23A's turn rate capability at 40,000 feet approaches 18 degrees per second across a wide range of Mach numbers. This exceptional high-altitude agility, combined with the aircraft's supercruise capability, would have provided a unique tactical advantage in beyond visual range engagements where the F-23A could maneuver aggressively while maintaining supersonic speed without afterburner.

The performance data reveals interesting characteristics that emerged from our modeling approach. The slight degradation in G-loading capability at 20,000 feet compared to both sea level and 40,000 feet reflects the challenging aerodynamic environment of the mid-altitude regime, where neither low-altitude density nor high-altitude efficiency advantages are fully realized. This characteristic actually aligns well with the known behavior of large-wing, high-aspect-ratio fighters and validates our parametric modeling approach.

Performance Characteristics and SFM Trade-offs

Our achieved performance envelope demonstrates that the F-23A SFM delivers exceptional high-speed performance at all altitudes, with particularly impressive capabilities where the aircraft's design advantages become most apparent. The supercruise performance modeled in our SFM reflects the YF-23's documented advantages in this regime, allowing sustained supersonic flight without afterburner across a wide range of altitudes and speeds that surpasses most contemporary fighters.

However, the SFM's limitations meant that certain aspects of the aircraft's performance had to be simplified or approximated. The YF-23's documented high angle of attack capabilities, for instance, are difficult to represent accurately within the SFM framework due to its simplified approach to post-stall aerodynamics and departure characteristics. Similarly, the subtle but important differences in control authority and handling qualities that would have distinguished the F-23A from its contemporaries are challenging to capture within the SFM's relatively rigid structure.

The performance charts reveal areas where the SFM's limitations forced compromises. The variations in achieved versus desired G-loading at different altitudes and Mach numbers reflect the difficulty of tuning the SFM's lookup tables to provide consistent performance across the entire flight envelope. Each data point required careful balancing to ensure that improvements in one area didn't create unrealistic penalties elsewhere—a process that consumed countless hours of flight testing and iterative refinement.

Looking Forward: Enhanced Flight Model Development

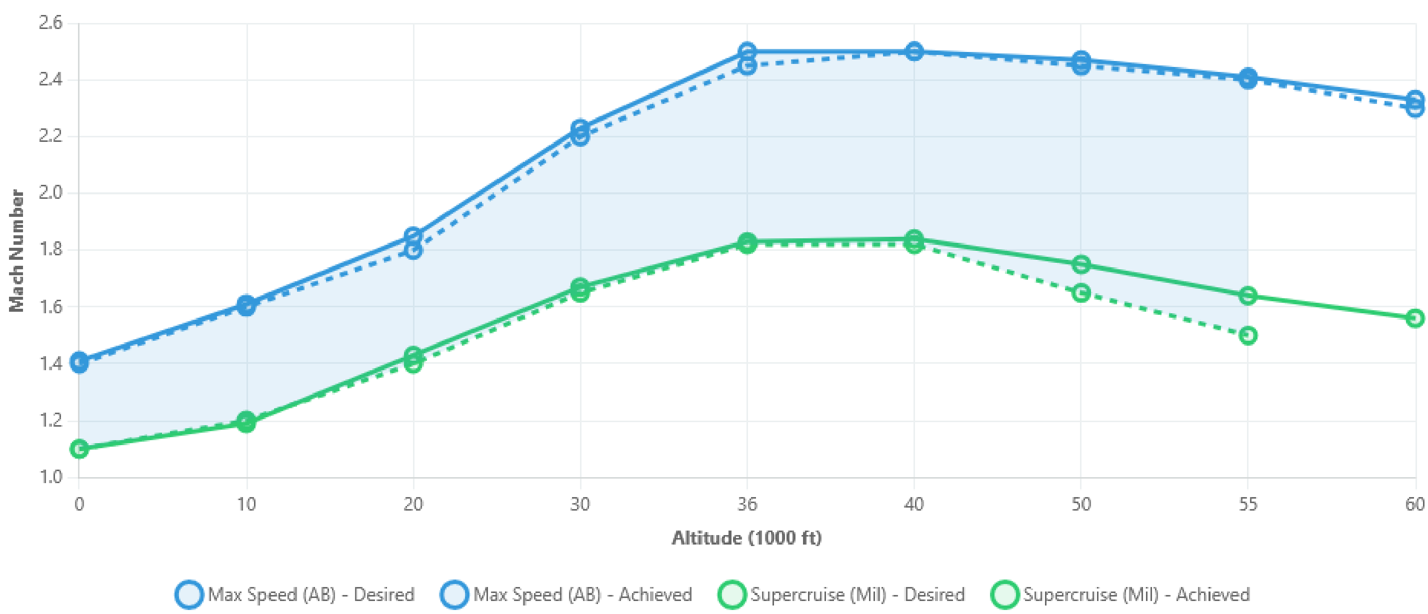
When we transition to the External Flight Model (EFM) for our F-23A Block 30 release, we anticipate being able to address many of these limitations. The EFM's more sophisticated approach to flight dynamics modeling will allow us to properly implement the high angle of attack performance that the YF-23 demonstrated, as well as the subtle handling characteristics that would have made the F-23A unique among fifth-generation fighters.

The EFM will also enable us to model the aircraft's flight control system behavior more accurately, including the sophisticated flight control laws that would have been necessary to manage the F-23A's inherent aerodynamic characteristics. This includes the aircraft's behavior in post-stall flight regimes, its departure and recovery characteristics, and the nuanced interaction between the pilot and the flight control system that defines the handling qualities of modern fighter aircraft.

We look forward to delivering to you a unique and exciting experience of the "what if" scenario of the F-23A having been brought to service—first through our carefully crafted SFM that captures the essence of this remarkable design, and eventually through an EFM that will allow us to explore the full potential of what might have been one of the most capable fighters ever designed.

Chart 1: Flight Envelope Performance

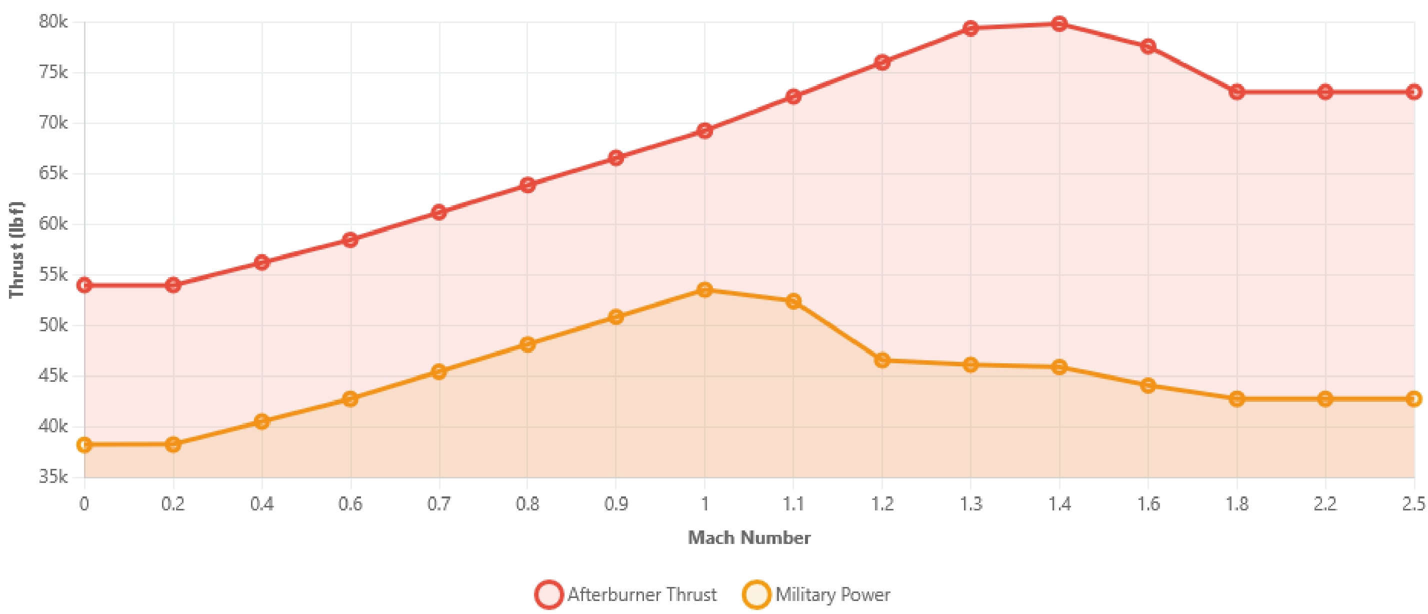
Maximum Speed and Supercruise Capability vs Altitude



Demonstrates high-altitude optimization with peak supercruise at 40,000+ feet

Chart 2: Sea Level Thrust Characteristics

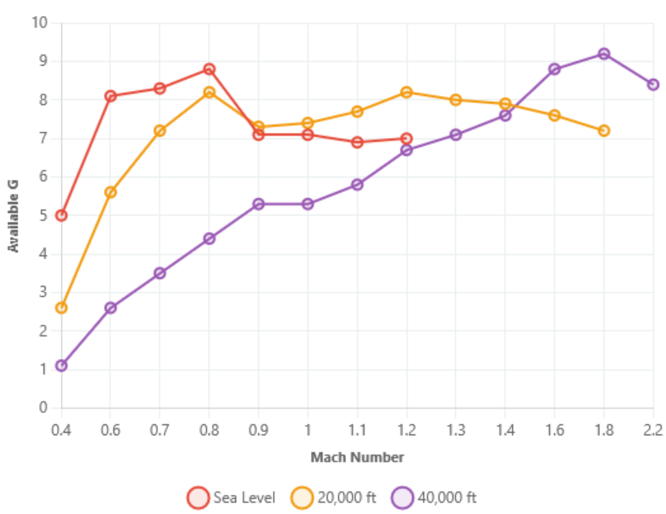
Engine Thrust Output vs Mach Number



F119-class powerplant with peak AB thrust approaching 80,000 lbf

Chart 3: G-Loading Performance

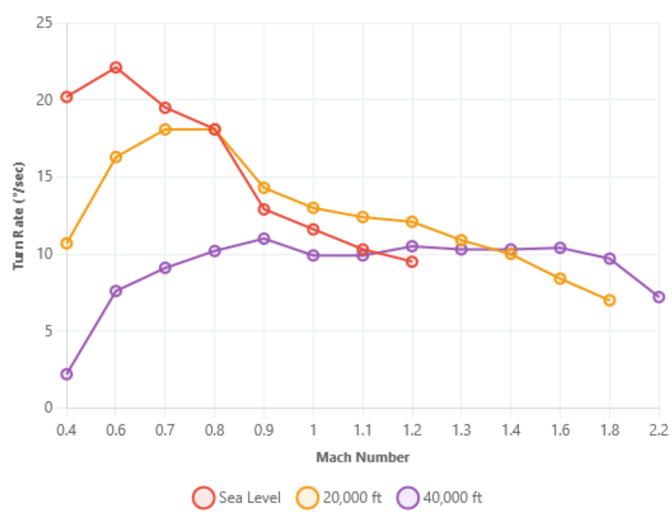
Available Load Factor vs Mach Number



Superior high-altitude maneuvering capability

Chart 4: Turn Rate Performance

Sustained Turn Rate vs Mach Number



Exceptional agility at 40,000 feet altitude

Credits

ThunderStruck Simulations

Brad Fischer "Smash" - Lead Developer and code

Mr. Benevolent Canard – Display Lead and code

Spurts – Flight Model lead

Nuffbutter – Artist and Liveries

Jeppef - Photography

Lukainaudi – Rearming Screen Art and UFCD design assistance

ziggysweet_soul_dust – UFCD loading Screen design

External Contributors 3D Artwork and Textures: HiMASTERS Art

All commissioned artwork and intellectual property rights belong to Brad Fischer

HiMASTERS website: https://himasters.art/

We would like to recognize all who have paved the way in the modding community before us. It is because of them that we are able to bring this mod to you. We would also like to thank those who have helped us on discord and various forums. This list is far from complete but notable names are:

- Grinnelli
- Loki_v2
- Kinkkujuustovoileipa
- Hayds_93
- Panic
- CubanAce